

formative-assessments\formative-week-4.py

```

1  '''
2  next problem
3  '''
4      #find the slope of the line with points:
5
6      (4, 3)
7      (0, (-4))
8
9       $m = (y^2 - y^1) / (x^2 - x^1)$ 
10      $m = ((-4) - 3) / (0 - 4)$ 
11      $m = (-7) / (-4)$ 
12      $[m = 7 / 4]$ 
13
14  '''
15  next problem
16  '''
17      #find the y-intercept and slope of:
18
19       $y = mx + b$ 
20
21       $(-4x) + 3y = (-2)$ 
22       $+ 4x \quad \quad + 4x$ 
23  #-----
24       $3y = 4x + (-2)$ 
25       $/ 3 \quad \quad / 3$ 
26  #-----
27      #slope-intercept form →
28       $y = (4 / 3)x + (-2 / 3)$ 
29      #slope →
30       $[m = (4 / 3)]$ 
31      #y-intercept →
32       $[b = (-2 / 3)]$ 
33
34  '''
35  next problem
36  '''
37      #find the slope of a line parallel to →
38
39       $x + 3y = 1$ 
40       $- x \quad \quad - x$ 
41  #-----
42       $3y = (-x) + 1$ 
43       $/ 3 \quad \quad / 3$ 
44  #-----
45       $y = (-1 / 3)x + (1 / 3)$ 
46
47       $[m\text{-parallel} = (-1 / 3)]$ 
48

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49     #find the slope of a line perpendicular
50     #reciprocal and with the opposite sign →
51
52     [m-perpendicular = (3 / 1)]
53
54     '''
55 next problem
56     '''
57     #find equation of line perpendicular to →
58
59     y = 5x + (-1)
60     y = (-1 / 5)x + (-1)
61
62     #and passes through point (3, 5)
63     #substitute x = 3 and y = 5 into y = (-1 / 5)x + b
64     #why is it b instead of the + (-1)? we're finding the new equation, not just the new y-
intercept
65
66     5 = (-1 / 5)(3) + b
67     5 = (-3 / 5) + b
68     b = 5 + (3 / 5)
69     b = (25 / 5) + (3 / 5)
70     b = 28 / 5
71     #equation of the line perpendicular and passing through (3, 5) →
72     y = (-1 / 5)x + (28 / 5)
73
74     #find equation of line parallel to →
75
76     y = 5x + (-1)
77     y = 5x + (-1)
78
79     #and passes through point (3, 5)
80
81
82
83
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90
91
```